

Task 4: Machine Learning Model Comparison

Logistic Regression vs Random Forest on Titanic Dataset

1. Introduction

Machine Learning model comparison is an essential step in predictive analytics. Different algorithms may perform differently depending on data structure, feature relationships, and complexity. The objective of this task is to build and compare two classification models Logistic Regression and Random Forest using the Titanic dataset and evaluate their performance using metrics such as Accuracy, Precision, Recall, and F1-score.

The Titanic dataset was selected because it provides a real-world classification problem: predicting whether a passenger survived based on demographic and socio-economic features.

2. Dataset Overview

The dataset contains 891 passenger records with features including:

- Pclass (Passenger Class)
- Sex
- Age
- SibSp (Siblings/Spouses aboard)
- Parch (Parents/Children aboard)
- Fare
- Embarked
- Survived (Target variable)

the dataset was loaded using pandas and prepared for modeling.

3. Data Preprocessing

3.1 Handling Missing Values

- Age missing values were filled using the median.
- Embarked missing values were filled using the mode.

This ensured no missing data during model training.

3.2 Encoding Categorical Variables

- Sex and Embarked were converted into numerical format.
- This allowed machine learning algorithms to process categorical data.

3.3 Feature and Target Separation

- X = All features except Survived
- y = Survived (Target)

3.4 Train-Test Split

The dataset was split using:

- 80% Training Data
- 20% Testing Data
- `random_state = 42` (for reproducibility)

4. Model 1: Logistic Regression

Logistic Regression is a linear classification algorithm that models the probability of a binary outcome.

- Accuracy: 0.8101
- Precision: 0.7857
- Recall: 0.7432
- F1 Score: 0.7639

Interpretation:

- The model correctly predicted survival status approximately 81% of the time.
- Precision indicates that when the model predicts survival, it is correct about 78% of the time.
- Recall indicates the model correctly identifies about 74% of actual survivors.
- The F1-score shows balanced performance between precision and recall.

5. Model 2: Random Forest

Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

- Accuracy: 0.8212
- Precision: 0.8088
- Recall: 0.7432
- F1 Score: 0.7746

Interpretation:

- Random Forest achieved higher accuracy than Logistic Regression.
- Precision improved to approximately 81%.
- Recall remained similar.
- F1-score increased, indicating better overall classification balance.

6. Confusion Matrix Analysis**Logistic Regression Confusion Matrix:**

[[90 15]

[19 55]]

Random Forest Confusion Matrix:

[[92 13]

[19 55]]

Analysis:

- Random Forest reduced False Positives (15 $\rightarrow$ 13).
- True Negatives improved (90 $\rightarrow$ 92).
- True Positives remained the same.

This indicates Random Forest handled non-linear relationships slightly better.

7. Feature Importance (Random Forest)

Most important features:

1. Sex
2. Fare
3. Age

Moderately important:

- Pclass

Less important:

- SibSp
- Parch
- Embarked

Interpretation:

- Gender was the strongest predictor of survival.
- Higher fare (economic status) increased survival probability.
- Age also played a significant role.

This aligns with historical evidence of evacuation priority given to women and higher-class passengers.

8. Model Comparison Summary

Metric	Logistic Regression	Random Forest
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Accuracy	81.01%	82.12%
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Precision	78.57%	80.88%
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Recall	74.32%	74.32%
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F1 Score	76.39%	77.46%
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Key Observations:

- Random Forest outperformed Logistic Regression in Accuracy, Precision, and F1-score.
- Logistic Regression remains simpler and more interpretable.
- Random Forest better captures complex feature interactions.

9. Conclusion

This study compared two machine learning models on the Titanic dataset. While both models performed well, Random Forest demonstrated superior predictive performance due to its ability to handle non-linear relationships and feature interactions.

However, Logistic Regression remains valuable for interpretability and computational efficiency.

The results demonstrate the importance of model comparison before selecting the final algorithm for deployment.